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5. VELOCITY OF SOUND IN AIR USING RESONANCE COLUMN

AIM

To determine the velocity of sound in air at room temperature using the resonance column.

APPARATUS REQUIRED

Resonance tube, three tuning forks of known frequencies, a rubber hammer, one thermometer, plumb line, set squares, water in a beaker.

FORMULA

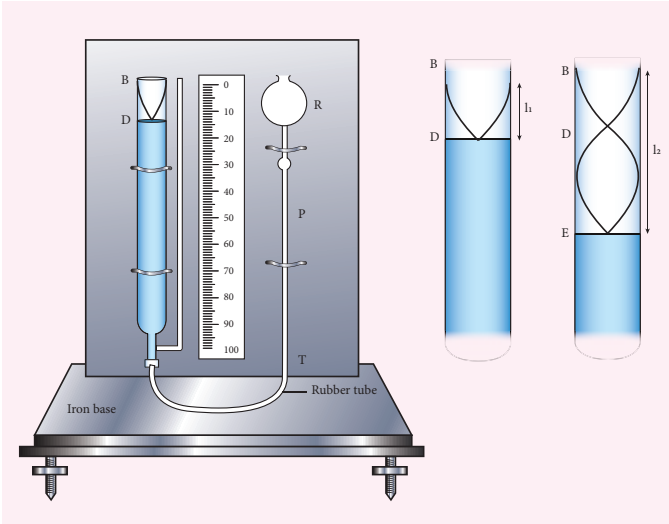
$V = 2v\left(l_2 - l_1\right)$ m s^{-1}

where V $\rightarrow$ Speed of sound in air (m s^{-1})

l_1 and l_2 $\rightarrow$ The length of the air column for the first and second resonance respectively (m)

$v \rightarrow$ Frequency of the tuning fork (Hz)

DIAGRAM



PROCEDURE

- The inner tube of the resonance column is lowered so that the length of air column inside the tube is very small.
- Take a tuning fork of known frequency and strike it with a rubber hammer. The tuning fork now produces longitudinal waves with a frequency equal to the natural frequency of the tuning fork.
- Place the vibrating tuning fork horizontally above the tube. Sound waves pass down the total tube and reflect back at the water surface.
- Now, raise the tube and the tuning fork until a maximum sound is heard.
- Measure the length of air column at this position. This is taken as the first resonating length, l_1